

MIMIC-CKD: A Large-Scale Benchmark for Comprehensive Survival Analysis in Chronic Kidney Disease

Abstract

Affecting more than 1 in 7 American adults, the prevalence of Chronic Kidney Disease (CKD) cannot be understated. Modeling CKD progression is crucial in identifying patients who are at high risk for end stage renal disease (ESRD), a disease with high mortality rates. However, existing models often make unrealistic assumptions such as time invariance, complete patient feature monitoring, and a reliance on pre-defined measures such as eGFR. Complete and standardized data in realistic hospital settings are rare. Currently, no CKD progression benchmarks exist to test model performance under imperfect settings. To the best of our knowledge, we construct the first large-scale CKD survival analysis benchmark dataset extracted from the large MIMIC-IV dataset with over 10,000 patients, 50,000 visits, and design several benchmark scenarios to better understand how survival analysis models perform across different feature sets such as time variant, invariant, and across a wide range of features. In our benchmarking, we show that the incorporation of time variant features directly improves model performance by upwards of 20%, and that exploring alternative features can boost performance upwards of an additional 4%.

Keywords: Deep learning, chronic kidney disease, heterogenous data

Data and Code Availability We use MIMIC-IV (Johnson et al., 2023) for our CKD benchmark construction. We offer our initial loader and implementation through the submitted code files.

Institutional Review Board (IRB) No IRB was required as we did not run any human experiments.

1. Introduction

Chronic kidney disease (CKD) affects over 1 in 7 Americans (Centers for Disease Control and Prevention, 2023) and is frequently associated with cardio-

vascular complications including stroke and heart disease (Said and Hernandez, 2014). CKD can progress to end-stage renal disease (ESRD), which has a five-year survival rate below 50%. While early intervention is critical for preventing ESRD (Levin and Stevens, 2011), predicting CKD progression remains challenging due to its ambiguous definition and often asymptomatic nature (Zhong et al., 2017). To identify reliable biomarkers of disease progression, various predictive models have been trained for CKD. In mining electronic health records (EHR) to better understand associated features with CKD, Chicco et al. (2021) find highly correlated biomarkers like age, eGFR, and creatinine, but more remarkably find that other temporally-relevant conditions like hypertension, smoking, and diabetes progression directly affect CKD. The significance of temporal features aligns with evidence that CKD risk increases with age (Hill et al., 2016). This temporal dimension is particularly relevant in clinical settings, where kidney function is monitored through recurring blood and urine tests, especially for diabetes-related complications (Chen et al., 2019). However, the varying importance of different features across studies highlights a critical gap between controlled research environments and real-world clinical scenarios, where patient data is often heterogeneous and incomplete across time points (Wu et al., 2024b) as depicted in Figure 1.

While many studies have performed CKD survival analysis, their datasets are either private (Fatin et al., 2023), have been conducted on a very small time-invariant scale (i.e less than 1,000 patients) (Swain et al., 2023), or fail to consider a wider range of features available within EHR.

As such, it is crucial to build a benchmark that leverages a more comprehensive set of features and outcomes. We construct a new dataset analyzing CKD progression from MIMIC-IV (Johnson et al., 2023).

To the best of our knowledge, our benchmark is the first to:

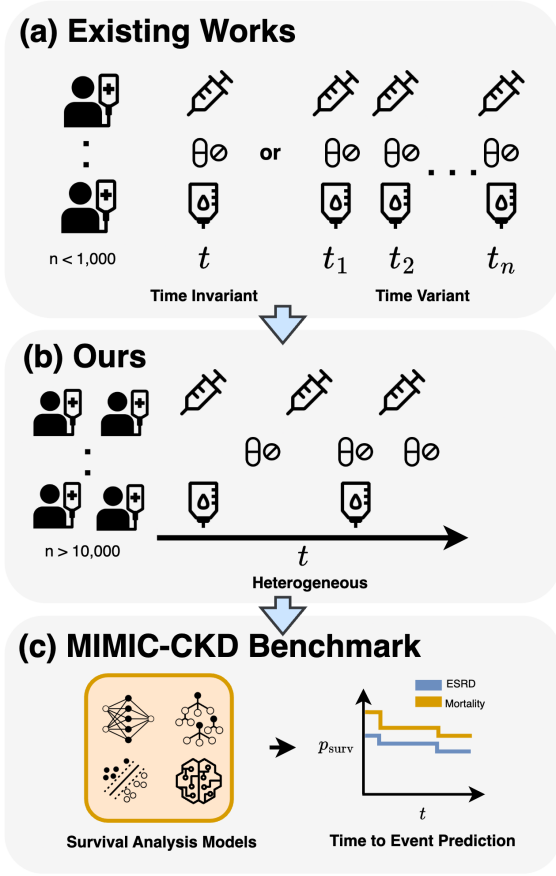


Figure 1: Patient profiles are often heterogeneous and complex in clinical settings. They vary considerably in the number of available features and time points of data collection. Leveraging this information to predict CKD progression is nontrivial. (a) In existing works surrounding CKD-ESRD time to event prediction, we observe that many ignore this environment and either assume time invariance (Swain et al., 2023) or uniform monitoring of lab events (Chen et al., 2019) across a small set of publicly available patient data. (b) Our benchmark explores the challenge of incorporating features that are not only missing across different patient profiles, but also across different time points among a much larger set of patients. (c) We explore the performance of various existing cited survival analysis models in our MIMIC-CKD benchmark.

We benchmark our dataset MIMIC-CKD across a wide variety of methods, including deep neural network survival models that can encode nonlinear relationships between the covariates and predicted outcomes such as DeepHit (Lee et al., 2018), Dynamic DeepHit (Lee et al., 2019), RNN-Surv (Giunchiglia et al., 2018), Hazard Transformer (Oliver et al., 2024), as well as more traditional techniques such as Cox models (Cox, 1972), gradient boosted machines (Chen et al., 2013), and survival random forests (Pickett et al., 2021). We observe that deep models on average outperform their traditional counterparts and scalably incorporate more features, improving performance upwards of 20%.

2. Methodology

2.1. Problem Formulation

Time-to-event analysis or survival analysis studies the probability of an event occurring by a given time. We note that this formulation is not unique to the medical field (Friedman et al., 2015), and has been applied in business (Morrison, 2004) and reliability engineering (Peña and Hollander, 2004). Nonetheless, within the medical space and typical EHR systems (Kline et al., 2022), there exist three main data scenarios as depicted in Figure 1, each with an additional layer of complexity: time-invariant, time-varying, and heterogeneous.

Time invariant and variant eGFR features. For the time invariant and time variant settings, we use the widely-established estimated glomerular filtration rate (eGFR) as our input biomarker, calculated using the CKD-EPI 2021 equation (Omuse et al., 2024):

$$\text{eGFR} = 142 \times \min\left(\frac{\text{Scr}}{\kappa}, 1\right)^\alpha \times \max\left(\frac{\text{Scr}}{\kappa}, 1\right)^{-1.2} \times 0.9938^{\text{Age}} \times (1.012)^{I[\text{Female}]}$$
(1)

where Scr is serum creatinine (mg/dL), and κ and α are gender-specific parameters.

Raw Features. We also explore time varying the raw features of the equation above as direct inputs rather than weighing it by its previously estimated formulation and compare the predictive performance in Table 3.

We further describe the three settings in Figure 1 explored by our benchmark in the subsections below.

- Include a large-scale cohort of over 10,000 patients across 50,000 clinical visits
- Capture realistic patient trajectories through temporally heterogeneous clinical features

2.1.1. TIME INVARIANT

In the time-invariant setting, we assume each patient has a snapshot of their profile, regardless of time. In this case, our survival data

$$\mathcal{D} = \{(\mathbf{x}_i, t_i, e_i)\}_{i=1}^n \quad (2)$$

is composed of three components where $\mathbf{x}_i \in \mathbb{R}^d$ with d the number of features, $t_i \in \mathbb{R}^+$, and $e_i \in \{0, 1, \dots, |E|\}$ where E is the set of all competing events (i.e., ESRD, mortality, etc.). When $e_i = 0$, the patient is considered to be right-censored.

In this scenario, the goal is to estimate the hazard function $h(t)$ with our model $f(t)$:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} \quad (3)$$

Here, $h(t)$ represents the instantaneous rate of event occurrence at time t , given that the event has not occurred before time t . More specifically, it is the probability that a patient experiences ESRD or death, given they survive up to that time point t . In some sense, we are essentially estimating the probability mass function (PMF) of the time-to-event:

$$f(t|\mathbf{x}) = P(T = t|\mathbf{x}) \quad (4)$$

2.1.2. TIME VARYING

However, as suggested in Section 1, most patients were hospitalized multiple times over the course of observation for this dataset. Thus, their profiles consist of multiple snapshots, which is commonly known as time-varying survival analysis (Zhang et al., 2018). Again, we define our dataset of triples of data, time of event, and event types:

$$\mathcal{D} = \{(\mathbf{x}_i, t_i, e_i)\}_{i=1}^n \quad (5)$$

where $\mathbf{x}_i \in \mathbb{R}^{s_i \times d}$, s_i is the number of time points observed for individual i , d is the feature dimension, and $t_i \in \mathbb{R}^+$. Note that the only difference here is that we have added a sequential component in the data $\mathbf{x}_i$.

2.1.3. HETEROGENEOUS

In the heterogeneous setting, we build upon the time-varying framework but account for missing features across different time points. Our dataset structure remains similar: $\mathcal{D} = \{(\mathbf{x}_i, t_i, e_i)\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^{s_i \times d}$, with s_i representing the number of observed

time points for individual i , and d is the feature dimension. However, unlike the time-varying setting, not all features are observed at each time point, resulting in a sparse representation. Formally, we can represent this as a mask $\mathbf{m}_i \in \{0, 1\}^{s_i \times d}$ where $m_{i,j,k} = 0$ indicates that feature $k \in \{1, \dots, d\}$ at time point $j \in \{1, \dots, s_i\}$ for individual i is missing.

In our specific implementation, each input representation for patient i at time j includes 2 types of features, eGFR and Hemoglobin with 4 columns to represent missing features: $\mathbf{x}_{i,j} = (\text{eGFR}_{i,j}, I_{\text{eGFR},i,j}^{(m)}, P_{i,j}, I_{P,i,j}^{(m)}, A_{i,j}, I_{A,i,j}^{(m)})$ where $I_{\text{eGFR},i,j}^{(m)} \in \{0, 1\}$ indicates that eGFR_{*i,j*} is missing when equal to 1 (similarly for $I_{P,i,j}^{(m)}$ and $I_{A,i,j}^{(m)}$). Here, $P_{i,j}$ represents proteinuria measurement for patient i at time j , $A_{i,j}$ represents serum albumin level for patient i at time j , and $I_{\cdot,i,j}^{(m)}$ denotes the missing data indicator with superscript (m) for "missing".

Models designed for this setting must account for both the temporal evolution of patient states and the informativeness of the missing pattern itself, as the absence of certain measurements may carry clinical significance (e.g., tests not ordered because the clinician deemed them unnecessary based on other observations). Nonetheless, in our benchmark, we note that this representation is not fixed, and should be explored in future work.

2.2. Benchmarked Models

In this subsection, we introduce each model evaluated in our survival analysis experiments across the different data scenarios described above.

2.2.1. TRADITIONAL SURVIVAL ANALYSIS MODELS

We include two parametric/nonparametric baseline models, implemented via the lifelines Python package (Davidson-Pilon, 2019).

Cox Proportional Hazards Model (Cox, 1972) models the hazard function as:

$$\lambda(t | \mathbf{x}) = \lambda_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}) \quad (6)$$

where $\lambda_0(t)$ is the baseline hazard function and $\boldsymbol{\beta}$ represents the coefficients that quantify the effect of covariates $\mathbf{x}$ on the hazard. The model is trained by maximizing the partial likelihood:

$$L(\boldsymbol{\beta}) = \prod_{i:e_i=1} \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_i)}{\sum_{j:t_j \geq t_i} \exp(\boldsymbol{\beta}^\top \mathbf{x}_j)} \quad (7)$$

We use the lifeline implementation for time invariant and time variant scenarios.

Weibull Accelerated Failure Time (AFT) model (Anderson, 1991) parameterizes the survival function as:

$$S(t | \mathbf{x}) = \exp \left(- \left(\frac{t}{\lambda(\mathbf{x})} \right)^\rho \right) \quad (8)$$

where $\lambda(\mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta})$ is the scale parameter and ρ is the shape parameter. Implemented as `from lifelines import WeibullAFTFitter`, we estimate regression coefficients $\boldsymbol{\beta}$ and shape parameter ρ . Continuous covariates are standardized; categorical covariates are one-hot encoded.

2.2.2. DEEP SURVIVAL ANALYSIS MODELS

Recent years have seen a surge in deep learning approaches for survival analysis, capable of handling complex data structures and relationships. All deep architectures consume time-indexed feature tensors $x_i \in \mathbb{R}^{s_i \times d}$ as described in the experimental settings.

DeepSurv (Katzman et al., 2018) is a feed-forward implementation of the Cox model with fully connected layers. We modify the original implementation to handle time-varying inputs by stacking each visit as a separate "instance" and encoding time-to-event as an additional feature. The network has three hidden layers (ReLU activations, size 128) followed by a linear output layer representing the log-hazard. We use Cox partial log-likelihood as the loss.

DeepHit (Lee et al., 2018) directly estimates the joint distribution of survival times and competing events without assuming any specific form for the underlying stochastic process. It employs a neural network architecture with a shared subnetwork and cause-specific subnetworks to model the discrete probability mass function:

$$P(T = t, E = k | \mathbf{x}) = y_{t,k} \quad (9)$$

where $y_{t,k}$ is the output of the network for time t and event k . The model is trained using a composite loss function:

$$L_{total} = L_1 + \alpha L_2 \quad (10)$$

where L_1 is the negative log-likelihood term accounting for right censoring:

$$L_1 = - \sum_{i=1}^n \left[\mathbf{1}_{(e_i \neq 0)} \log(y_{t_i, e_i}) + \mathbf{1}_{(e_i = 0)} \log \left(1 - \sum_{k=1}^K \sum_{t \leq t_i} y_{t,k} \right) \right] \quad (11)$$

and L_2 is a ranking loss that encourages proper ordering of events:

$$L_2 = \sum_{k=1}^K \sum_{i,j: e_i=k, t_i < t_j} \sigma \left(\hat{F}_k(t_i | \mathbf{x}_j) - \hat{F}_k(t_i | \mathbf{x}_i) \right) \quad (12)$$

Here, $\hat{F}_k(t | \mathbf{x})$ is the estimated cumulative incidence function for event k up to time t given covariates $\mathbf{x}$, and σ is a smoothing function. This composite loss formulation has inspired many subsequent deep survival models.

Dynamic-DeepHit (Lee et al., 2020) extends DeepHit to irregularly sampled time series. At each timestep we feed $\{\mathbf{x}_{i,j}, m_{i,j}, \Delta t_{i,j}\}$ where $\Delta t_{i,j} = t_{i,j} - t_{i,j-1}$. The architecture comprises a shared MLP encoder, a bi-directional LSTM layer, and cause-specific MLP heads that output discrete hazard probabilities over pre-specified time bins. The model uses the combined log-likelihood and ranking loss from DeepHit, tuning coefficient $\alpha \in [0.1, 1.0]$.

RNN-Surv (Martinez et al., 2020) uses gated recurrent units (GRUs) to model temporal dependencies and outputs risk scores for each patient at each visit. The architecture includes an input embedding layer projecting $\mathbb{R}^d \rightarrow \mathbb{R}^{64}$, two-layer GRU (hidden size 64), and a time-to-event decoder mapping the last hidden state to a scalar risk score. Different from Dynamic-DeepHit, RNN-Surv model employs the traditional negative Cox partial log-likelihood on final risk scores.

LogisticHazard implements a discrete-time survival model by framing event hazards in logistic form. Time is first divided into a fixed number of intervals, and for each interval the model learns the log-odds of "failure" versus "survival" in that window. Under the hood it's a neural network whose final layer outputs one logit per time interval; training minimizes the negative log-likelihood of the observed interval-censored outcomes. This approach marries classic discrete-time hazard modeling with the flexibility of

deep nets, allowing rich feature representations, non-linear covariate effects, and end-to-end learning of survival curves directly from data.

Hazard Transformer. Inspired by the original Transformer model (Vaswani et al., 2017), the Hazard Transformer (Oliver et al., 2024) replaces RNNs with the self-attention mechanism. It consists of positional encoding added to each $\mathbf{x}_{i,j}$, two Transformer encoder blocks (4 heads, hidden size 64; feedforward sublayers size 128), and a final linear layer producing discrete hazard probabilities over pre-binned intervals. The model uses cross-entropy-based ranking loss inspired by the DeepHit framework, with dropout 0.2 on attention and feedforward layers.

2.2.3. ENSEMBLE-LEARNING-BASED MODELS

We evaluated two tree-based survival learners from ‘scikit-survival’:

Gradient Boosting Survival Analysis (GBSA) (Chen et al., 2013) offers a powerful nonparametric alternative that can capture complex, non-linear relationships without imposing restrictive assumptions on the hazard function form. The objective function approximates the Cox model through additive ensemble learning:

$$L = \sum_i \ell(y_i, \hat{y}_i) + \Omega(f) \quad (13)$$

where ℓ is the negative log partial likelihood and $\Omega(f)$ is regularization. For brevity’s sake, we defer to the literature for their exact details.

Random Survival Forest (SRF) (Pickett et al., 2021; Ishwaran et al., 2008) extends random forests to survival data. Implemented as `sksurv.ensemble.RandomSurvivalForest`, it uses Harrell’s concordance index (C-index) on out-of-bag samples to guide hyperparameter selection. Again, we defer to the literature for their exact implementation.

FastSurvivalSVM adapts support vector machines to right-censored survival data by optimizing a convex surrogate for the concordance index. It learns a linear ranking function over feature space that pushes subjects with earlier events to have higher “risk scores” than those with later or censored times, subject to margin constraints and regularization.

2.3. Model Training and Optimization

Hyperparameter Tuning for Deep Models: For RNN-based and Transformer architectures, we per-

form Bayesian hyperparameter optimization using Optuna with 10 trials and 50 epochs per trial. Learning rates are tuned in log-uniform range $[10^{-5}, 10^{-2}]$, dropout rates in $[0.1, 0.5]$, and weight decay in $[10^{-6}, 10^{-3}]$.

Ensemble Model Tuning: For GBSA and SRF, we perform grid search over number of trees, learning rates, max depths, and minimum samples split parameters.

Data Preprocessing: Categorical variables are one-hot encoded, and we use an 80/20 train/test split by patient ID to ensure generalization capability. Traditional models use default optimization routines, deep models use Adam with early stopping, and ensemble models use validation-based selection.

3. Results

	Non-ESRD	ESRD
Patient Demographics		
# of Patients	2,790	7,389
Age (years)	74.8 ± 12.5	71.8 ± 13.7
Male	55.7%	57.5%
Hypertension	50.8%	58.4%
Diabetes mellitus	20.3%	31.0%
ACE inhibitor/ARB use	34.2%	43.6%
# of admissions	6,773	47,441
Laboratory Values		
Serum creatinine (mg/dL)	2.1 ± 2.0	2.3 ± 1.7
eGFR (mL/min/1.73m ²)	46.4 ± 24.4	41.0 ± 23.4
Proteinuria (g/24h)*	0.8 [2.3]	0.5 [4.3]

Table 1: Baseline Characteristics of CKD Stage 3-5 Patients by ESRD Status. This analysis focuses exclusively on patients with chronic kidney disease stages 3-5 (n=10,179), stratified by whether they developed end-stage renal disease (ESRD) during follow-up. Continuous variables are presented as mean $\pm$ standard deviation, except for proteinuria which is presented as median with interquartile range due to its skewed distribution. We investigate a wide variety of useful features defined by (Zhong et al., 2017) and more recently (Dana et al., 2024). We investigate these features across time. We also investigate features that may not necessarily be omnipresent across all time points.

3.1. Dataset Statistics

Given that ESRD risks are substantially higher in later CKD stages (Sud et al., 2014), we identified

10,179 patients with stage 3–5 CKD, of whom 7,389 progressed to ESRD. As shown in Table 1, ESRD patients were younger, had higher rates of diabetes and hypertension, more severe renal dysfunction, and increased use of renin–angiotensin inhibitors compared to non-progressors, establishing clinically meaningful baseline differences for model comparison.

Evaluation metrics. We benchmark our survival analysis models using three conventional metrics: concordance index (C-index) (Longato et al., 2020), Brier score (Assel et al., 2017), and area under the ROC curve (AUC) (Huang and Ling, 2005).

3.2. Time Invariant

We benchmark several traditional time-invariant models for CKD survival analysis as discussed in Section 2.2. Specifically, we evaluate traditional models (Cox proportional hazards, Weibull), ensemble methods (gradient boosted survival analysis or GBSA, survival random forests or SRF), and the neural network-based DeepSurv model.

Model	C-Index $\uparrow$	Brier score $\downarrow$	AUC $\uparrow$
Cox	0.517 $\pm$ 0.008	0.443 $\pm$ 0.001	0.527 $\pm$ 0.012
DeepSurv	0.523$\pm$0.002	0.475 $\pm$ 0.012	0.464 $\pm$ 0.002
GBSA	0.523 $\pm$ 0.011	0.329 $\pm$ 0.043	0.534 $\pm$ 0.016
SRF	0.517 $\pm$ 0.006	0.502 $\pm$ 0.004	0.527 $\pm$ 0.009
Weibull	0.521 $\pm$ 0.001	0.485 $\pm$ 0.007	0.535$\pm$0.001
LogisticHazard	0.514 $\pm$ 0.010	0.566 $\pm$ 0.008	0.496 $\pm$ 0.023
Fast Survival SVM	0.510 $\pm$ 0.007	0.252$\pm$0.004	0.485 $\pm$ 0.010

Table 2: Performance of Time-Invariant Models. Models use static patient snapshots without longitudinal information. Bold values indicate best performance for each metric.

More complex and less interpretable models do not provide any significant benefits in time invariant settings. At the cost of explainability, we see a small improvement in DeepSurv over the Cox model in C-index. However, the interpretable Weibull model is comparable and better depending on the metric. Nonetheless, traditional models like the Cox or Weibull model seem comparable in performance to the less interpretable SRF, GBSA, and DeepSurv models when considering both metrics.

3.3. Time Variant and Heterogeneous Scenarios

To understand the predictive capabilities of survival analysis models under different time variant settings, we benchmarked 4 models. Namely, we benchmark

the Cox model, and a variety of deep architectures such as Dynamic-DeepHit (Lee et al., 2019), Hazard Transformer (Oliver et al., 2024), and RNN-Surv (Martinez et al., 2020).

Time-Variant (Baseline)			
Model	C-Index $\uparrow$	Brier $\downarrow$	AUC $\uparrow$
Cox	0.545 $\pm$ 0.001	0.215 $\pm$ 0.002	0.539 $\pm$ 0.002
Dynamic-DeepHit	0.735$\pm$0.069	0.036$\pm$0.003	0.798$\pm$0.197
Hazard Transformer	0.500 $\pm$ 0.000	0.566 $\pm$ 0.007	0.500 $\pm$ 0.000
RNNSurv	0.640 $\pm$ 0.076	0.082 $\pm$ 0.088	0.648 $\pm$ 0.080
Logistic Hazard	0.514 $\pm$ 0.010	0.566 $\pm$ 0.008	0.496 $\pm$ 0.023
Time-Variant Raw			
Model	C-Index $\uparrow$	Brier $\downarrow$	AUC $\uparrow$
Cox	0.374 $\pm$ 0.004 $\downarrow$	0.926 $\pm$ 0.029 $\uparrow$	0.352 $\pm$ 0.004 $\downarrow$
Dynamic-DeepHit	0.780 $\pm$ 0.065 $\uparrow$	0.037 $\pm$ 0.002 =	0.830 $\pm$ 0.187 $\uparrow$
Hazard Transformer	0.500 $\pm$ 0.000 =	0.566 $\pm$ 0.007 =	0.500 $\pm$ 0.000 =
RNNSurv	0.668 $\pm$ 0.042 $\uparrow$	0.096 $\pm$ 0.057 =	0.681 $\pm$ 0.040 $\uparrow$
Logistic Hazard	0.507 $\pm$ 0.016 =	0.566 $\pm$ 0.007 =	0.513 $\pm$ 0.021 =
Heterogeneous Features			
Model	C-Index $\uparrow$	Brier $\downarrow$	AUC $\uparrow$
Cox	0.406 $\pm$ 0.003 $\downarrow$	0.294 $\pm$ 0.007 $\uparrow$	0.389 $\pm$ 0.003 $\downarrow$
Dynamic-DeepHit	0.761 $\pm$ 0.014 $\uparrow$	0.056 $\pm$ 0.000 $\uparrow$	0.492 $\pm$ 0.047 $\downarrow$
Hazard Transformer	0.500 $\pm$ 0.000 =	0.546 $\pm$ 0.000 $\downarrow$	0.500 $\pm$ 0.000 =
RNNSurv	0.647 $\pm$ 0.002 $\uparrow$	0.127 $\pm$ 0.043 $\uparrow$	0.651 $\pm$ 0.003 $\uparrow$
Logistic Hazard	0.518 $\pm$ 0.008 =	0.529 $\pm$ 0.027 $\uparrow$	0.526 $\pm$ 0.015 =

Table 3: Stacked Performance Comparison Across Settings. Time-Variant serves as baseline. Green indicates improvement (> 0.02 difference), red indicates degradation (> 0.02 difference), gray indicates no significant change (≤ 0.02 difference). For C-Index and AUC, higher is better ($\uparrow$); for Brier score, lower is better ($\downarrow$). Dynamic-DeepHit shows consistent strong performance across all settings and sees upwards of a 4% performance improvement in the raw setting.

Incorporating temporal features improves performance, but only when models can effectively leverage them. We observe that while the Cox model benefits from additional time points in the time-variant setting, the improvement is modest—approximately 3% increase in predictive performance. In contrast, deep architectures such as Dynamic-DeepHit and RNNSurv achieve substantial performance gains exceeding 20%. Notably, the transformer architecture underperforms across all metrics, yielding results worse than even time-invariant models. This poor performance of our Hazard Transformer aligns with existing literature, as transformers typically require advanced training techniques or extensive feature engineering to effectively model time-series data (Wen et al., 2023).

eGFR is a flawed composite feature. While eGFR is commonly used in clinical settings, its for-

mulation is not necessarily optimal. In fact, models can directly learn relationships between eGFR components, as evidenced by improved performance in Dynamic DeepHit and RNNSurv. In contrast, more interpretable and linear models like our Cox baseline show substantial performance declines. This observation makes sense because the eGFR equation essentially functions as a form of feature engineering, with explicitly defined weights for each component.

Current survival analysis models struggle to effectively utilize heterogeneous clinical features. Features like hemoglobin are commonly collected from CKD patients since anemia is a common CKD complication [Atkinson and Warady \(2018\)](#). However, Table 3 reveals that incorporating these heterogeneous features produces negligible improvements or actual performance decreases. This limitation represents an important challenge requiring further investigation in model architecture design and remains an active research area in healthcare AI systems.

While Dynamic DeepHit shows C-index improvements, suggesting better risk ranking across patient cohorts, its substantial drops in AUC and Brier scores indicate that heterogeneous features result in poorly calibrated models. Every model shows declining Brier scores, suggesting that while relative ranking predictions remain unchanged, incorporating heterogeneous inputs severely disrupts model calibration.

3.4. Feature Analysis

We run a feature analysis to better understand the importance of each input feature across both our raw and heterogeneous scenarios. In particular, we want to understand what features are driving each model’s predictions as shown in Figures 2 and 3.

Static features are still highly relevant for time-series predictive performance. Remarkably in Figure 2, our best performing model Dynamic DeepHit weighs static characteristics such as gender and age just as important as serum creatinine.

Missing features highly affect performance within our models. We include a missing feature indicator as part of our feature set in our heterogeneous case across different time points. In Figure 3, we observe that across all models our “missing” features are highly relevant in downstream predictions.

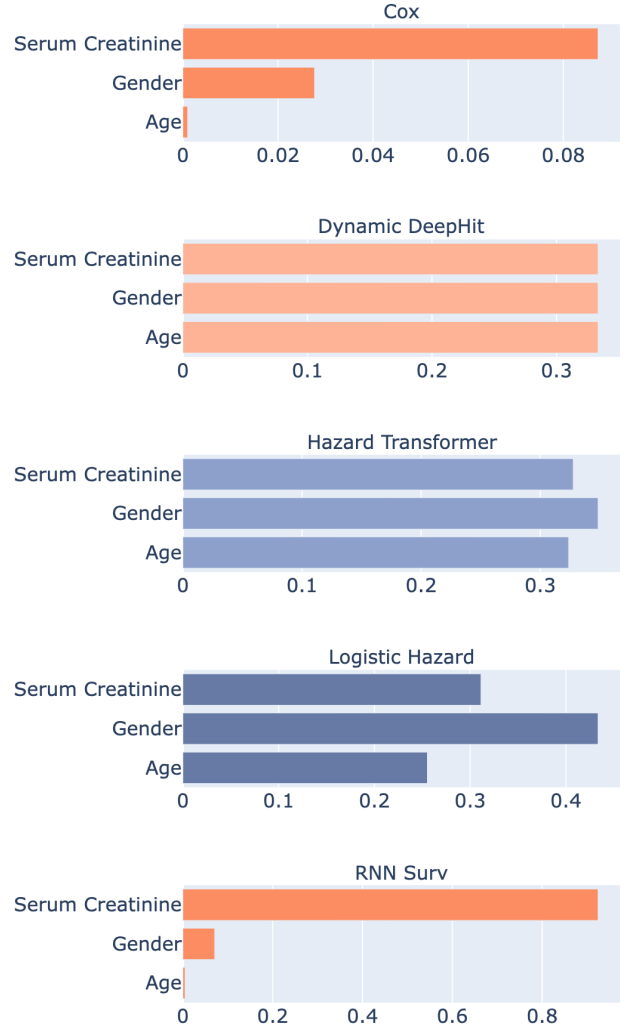


Figure 2: Feature Importance of the “Raw” Setting. We explore how directly inputting the individual components of the eGFR equation drives performance. Here, we plot our Cox model’s weight coefficients along with the gradient-based saliency maps ([Simonyan et al., 2014](#)) across our neural baselines. We observe that serum creatinine is highly relevant across all models, but age and gender varies depending on the model.

4. Discussion

Developing survival analysis models capable of handling the heterogeneous complexities of CKD progression prediction. Learning with missing modalities represents an active area of research ([Wu et al., 2024a](#)), particularly when features exhibit varying dimensionalities and structural char-

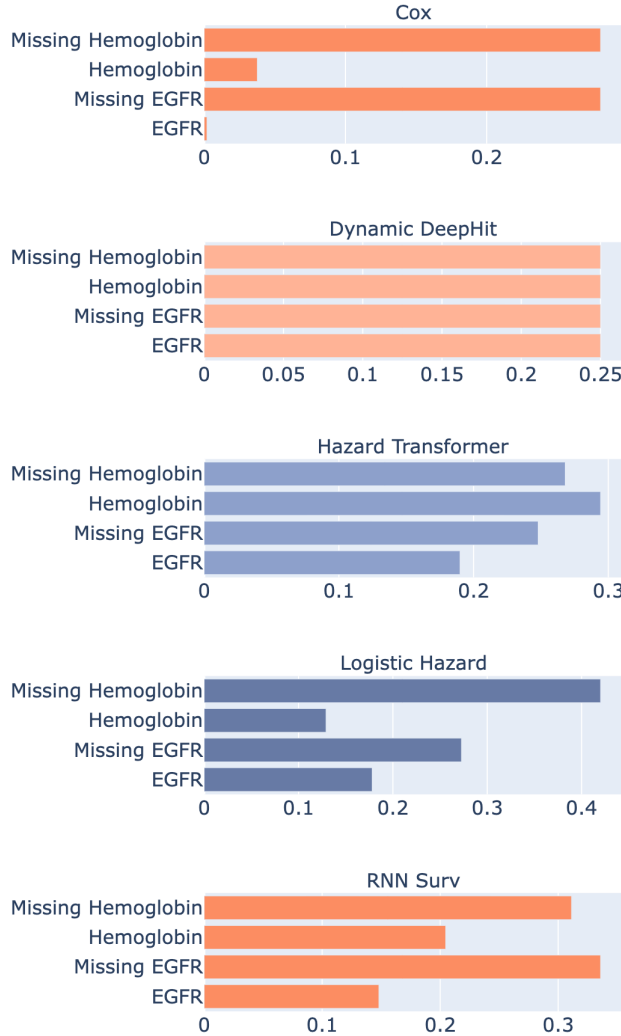


Figure 3: Feature Importance Within The Heterogeneous Setting. We explore how (missing) features affect downstream performance by running a feature analysis. Here, we plot our Cox model’s weight coefficients along with the gradient-based saliency maps (Simonyan et al., 2014) with respect to each input feature across our neural baselines. We observe that missing features have a massive effect on model predictions.

resents a critical advancement toward building more accurate and clinically realistic survival models.

Inclusion of competing events. There are many competing conflicts when it comes to predicting ESRD such as mortality (Hsu et al., 2017). While (Li et al., 2023) constructs a classification benchmark to predict the mortality of CKD patients, we note that it is crucial to consider both events in modeling as ESRD influences many other CKD-related complications, including mortality. Integrating these predicted complications would produce more realistic progression models where outcomes are highly multifaceted.

Multimodal Progression Models. MIMIC-IV (Johnson et al., 2023) contains substantially more features than our current benchmark incorporates, as shown by (Dana et al., 2024). Future work should systematically explore these features to understand their predictive value. Underexplored variables like social determinants of health (SDOH) (Chen et al., 2020) are particularly promising, as they may reveal critical differences in disease progression between patients who develop ESRD and those with stable kidney function.

Building robust healthcare predictive systems requires multimodal models that integrate diverse feature sets—including images (Johnson et al., 2019), text (Johnson et al., 2023), physiological signals (Jing et al., 2018), and demographic data (Pollard et al., 2018)—regardless of structural differences. Current CKD progression models typically ignore these features or rely on unrealistic imputation assumptions (Swain et al., 2023). Comprehensive CKD progression modeling ultimately requires features spanning multiple modalities.

acteristics. Despite these challenges, incorporating diverse modalities has demonstrated clear benefits, with studies showing improved performance in clinical prediction tasks including mortality and readmission forecasting (Wu et al., 2024b). Adapting these architectures—originally designed for handling missing modalities—to time-series survival analysis rep-

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